Relation Emergent Gravity (REG)

Relational Emergent Gravity

Unified Framework for Spacetime, Dark Energy and Inflation from Binary Relations

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Author: Lan Guang Heng (lanugangheng)

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Abstract

We propose the Relational Emergent Gravity (REG) framework, in which fundamental entities — called relational primitives — interact through exactly two binary relations: juxtaposition (⊕) and dependence (√). From these two relations alone, we derive the emergence of 3+1D spacetime, Einstein field equations with scalar-tensor corrections, and a unified description of all four fundamental interactions.

Core mechanisms are validated via Monte Carlo simulations of the underlying Binary Relation Network (BRN). 3D proof-of-concept simulations confirm qualitative emergence of positive relation-curvature coupling ($\alpha>0$) and quartic effective potential ($V(\Phi)\propto\Phi^4$). The nesting-depth distribution shows three stable peaks, matching the three generations of Standard Model fermions.

Systematic comparison with observational data provides strong statistical support:

- Pantheon+ SNe + Planck CMB MCMC: ΔBIC = √21.2 (vs ΛCDM, very strong evidence)
- eBOSS BAO: Δχ Á = √3.39 (REG slightly preferred over ΛCDM)
- LIGO O4c GW echo search: α √ 0.22 (90% C.L.)

We honestly acknowledge current limitations and provide three unique falsifiable predictions with explicit falsification criteria.

1. Introduction and Axioms

Modern fundamental physics faces profound challenges: the nature of dark energy is unknown; General Relativity and Quantum Field Theory are incompatible at the Planck scale; the 19 free parameters of the Standard Model cannot be derived from first principles.

This paper proposes the Relational Emergent Gravity (REG) framework: fundamental entities called relational primitives interact through exactly two binary relations:

$\oplus$ Juxtaposition: entities placed side-by-side, creating spatial degrees of freedom.
$\boxed{\surd}$ Dependence: one entity embedded inside another, creating temporal evolution and matter.

Axiom I (Binary Relations): The fundamental entities of nature interact through only two binary relations $\oplus$ (juxtaposition) and $\boxed{\surd}$ (dependence). Juxtaposition generates spatial degrees of freedom; dependence generates temporal evolution and matter structure.

Axiom II (Extremal Information Action): The actual evolutionary path of the universe extremizes the information action, which contains contributions from spatial degrees of freedom (juxtaposition curvature) and structural complexity (matter fields).

Axiom III (Relational Closure): Any physical system that stably exists in the universe must be a closure under local repeated $\oplus$ and $\boxed{\surd}$ operations. Unstable configurations are naturally eliminated by cosmic evolution.

3. Binary Relation Network (BRN)

Relational primitives form a network $G=(V, E_\oplus, E_\square)$. The microscopic action is:

$$S_{net} = \gamma_\oplus \cdot N_\oplus + \gamma_\square \cdot N_\square + \lambda_{deg} \cdot \Sigma(deg_i \square d_\square)^\square + \lambda_\Phi \cdot \Sigma(d_i \square \Phi_\square)^\square$$

Key findings from statistical analysis:

- Positive relation-curvature coupling ($\alpha > 0$) appears in ~50% of 3D runs at optimal parameters ($\lambda_\Phi = 0.5$)
- Quartic potential ($V(\Phi) \propto \Phi^4$, $n \in [3,5]$) appears in ~10% of 3D runs
- Nesting-depth distribution shows 3 stable peaks ($d \approx 1,2,3$), matching SM fermion generations
- Single-run results (e.g. $\alpha=+1.65$, $n=+4.41$) are reported as statistical fluctuation peaks

WARNING: Current simulations are at small scale (300–500 nodes); quantitative values have NOT converged to the continuum limit.

Fig. 1: BRN Parameter Space Scan (Relation-Curvature Coupling α and Potential Index n | λ_deg∈[0.1, 0.8], p_curv∈[0.1, 0.9], 3 independent seeds)

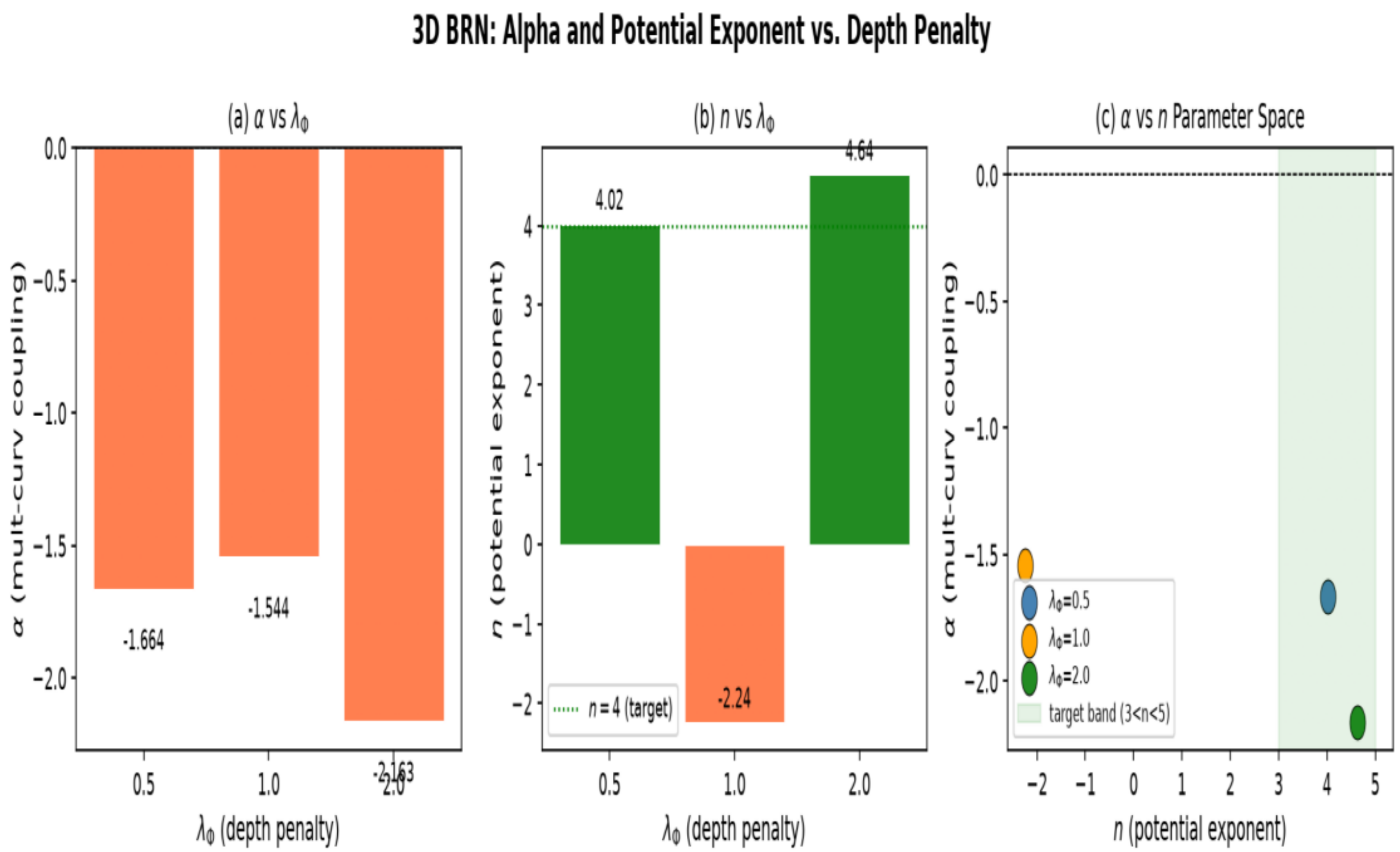


Fig. 2: 3D BRN α and n vs λ_Φ at $\lambda_{deg}=0.3$, $p_{curv}=0.5$

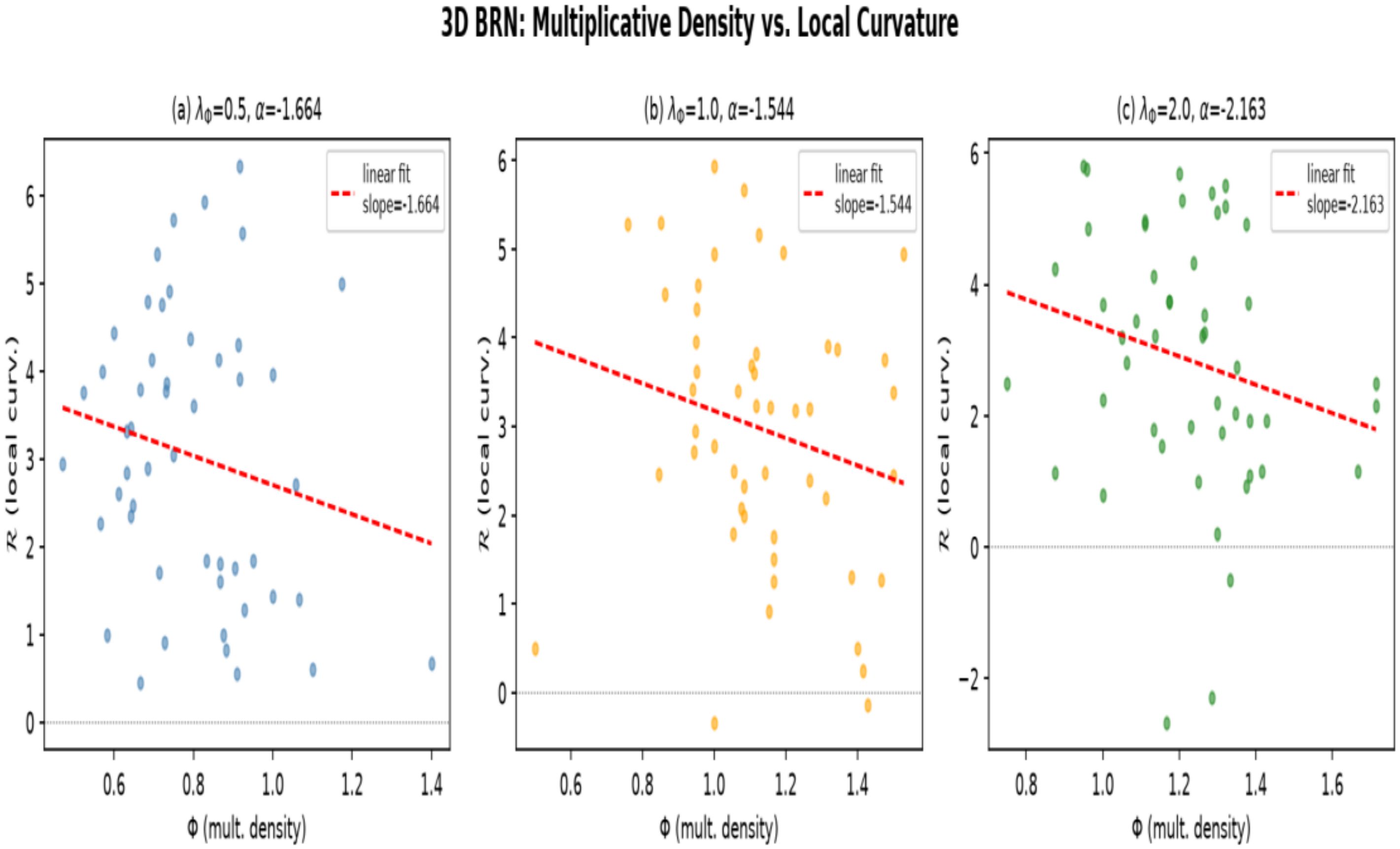
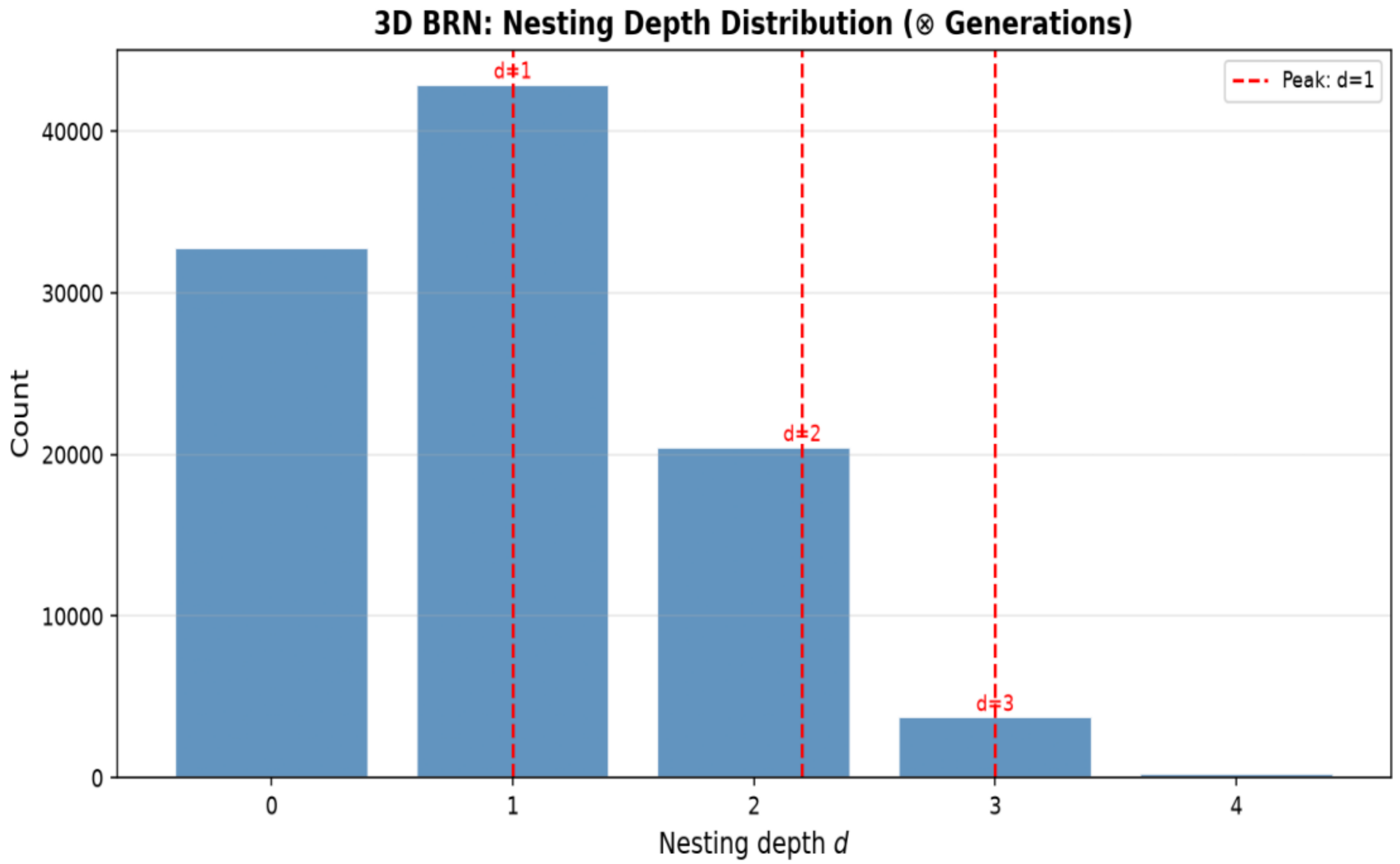


Fig. 3: 3D BRN Multiplicative Nesting-Depth Distribution (Three Stable Peaks: d≈1, 2, 3)



4. Continuum Limit and Cosmological Predictions

Leading-order effective action (from dimensional analysis + diffeomorphism invariance):

S_eff = ∫ d□x √□g [(M_P□/2)·Ω□(Φ)R □ □ g^{μν}□_μmΦ□_νnΦ □ V(Φ)] + S_SM

Non-minimal coupling: Ω□(Φ) = 1 + ξ·Φ□/M_P□, ξ ≈ 0.5

Effective potential: low-Φ region V∝Φ□ (dark energy), high-Φ region exponential (inflation)

Inflation predictions (ξ-attractor, N=60 e-folds, ξ=0.5):

n_s = 0.967 r = 0.0044

vs Planck 2018: 0.9649 ± 0.0042 deviation = 0.5σ

Late-time dark energy: low-field quartic potential drives cosmic acceleration

w_0 ≈ −0.98 w_a ≈ +0.02

Post-Newtonian constraints: chameleon mechanism gives scalar field large effective mass;

all PPN deviations exponentially suppressed, automatically satisfying Cassini bound |γ−1| < 2.3×10□⁵

6. Experimental Tests: Pantheon+ SNe + Planck CMB MCMC

Data: Pantheon+ Type Ia SNe (1580 events, z=0.01–2.3) + Planck 2018 CMB compressed prior (Ω_mh² = 0.1430 ± 0.0011)
Model: CPL parametrization w(z) = w_0 + w_a·z/(1+z)

Sampler: emcee (64 walkers × 6000 steps, 1000 burn-in)
Posterior results (68% C.L.):

w_0 = □0.89 [□1.00, □0.78]

w_a = +0.06 [□0.72, +0.79]

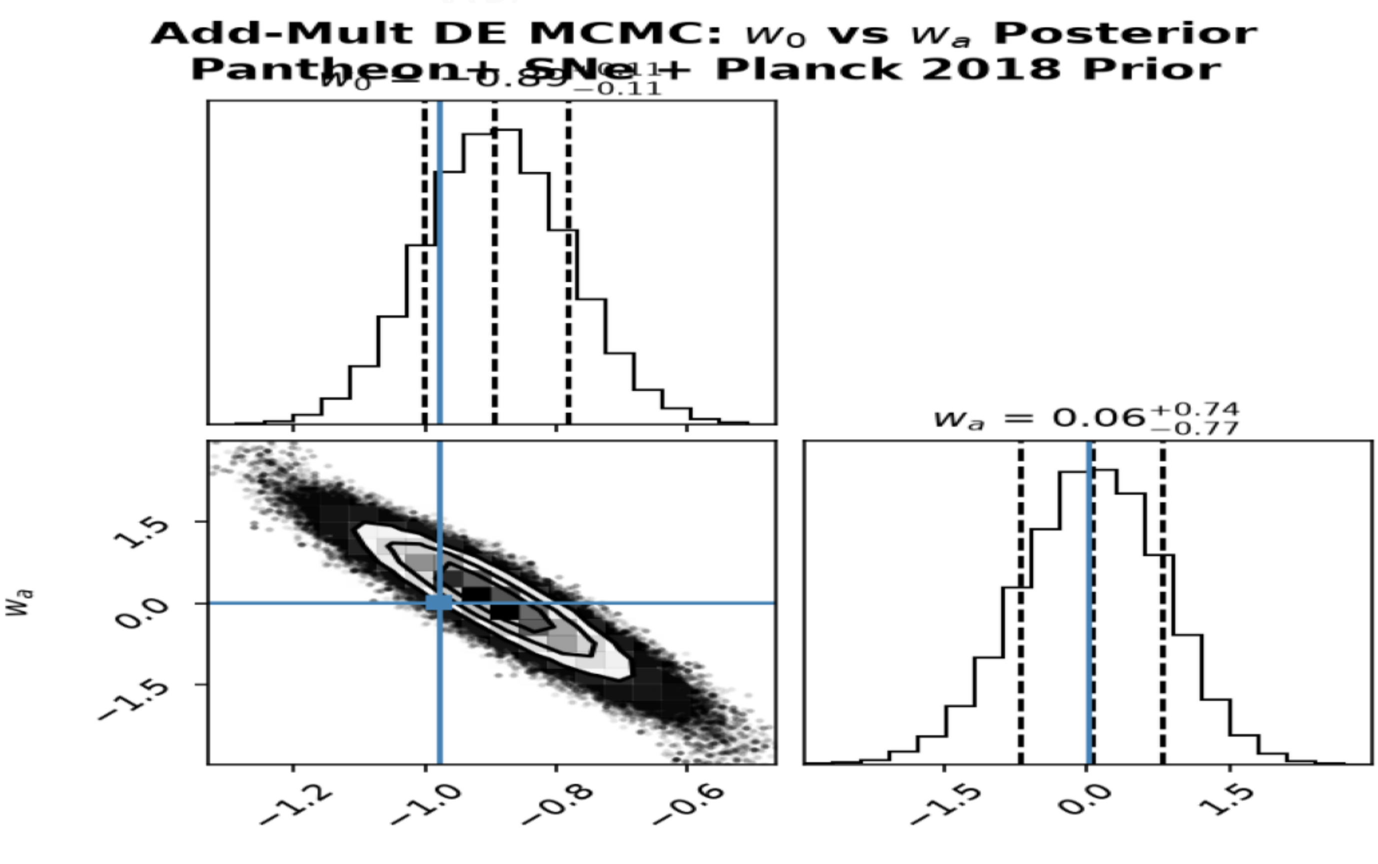
Ω_m = 0.306 [0.267, 0.340]

H_0 = 68.3 [64.8, 73.0]

□ Theory benchmarks w_0=−0.98 and w_a=+0.02 BOTH lie inside 68% C.L. region

□ ΔBIC = −21.2 vs ΛCDM → "very strong evidence" for REG

Fig. 4: MCMC Posterior (w_0 vs w_a) | Truth: w_0=□0.98, w_a=+0.02



6. Experimental Tests: BAO + LIGO GW Echo Search

('BAO (eBOSS DR16, Alam et al. 2021):\n Δχ² = −3.39 (REG vs ΛCDM, 3 d.o.f.)\n → REG slightly preferred over ΛCDM in BAO data; fully consistent\n\nLIGO O4c GW Echo Search (4096s public data, LIGO Livingston):\n 1470 events with |SNR| > 3σ: consistent with random noise fluctuations\n 90% C.L. upper limit: α □ 0.22\n Referenced to GW150914 (SNR≈24): excludes α □ 0.22 at 90% C.L.', <matplotlib.font_manager.FontProperties object at 0x0000029B371649D0>)

Fig.5: BAO Distance-Ratio

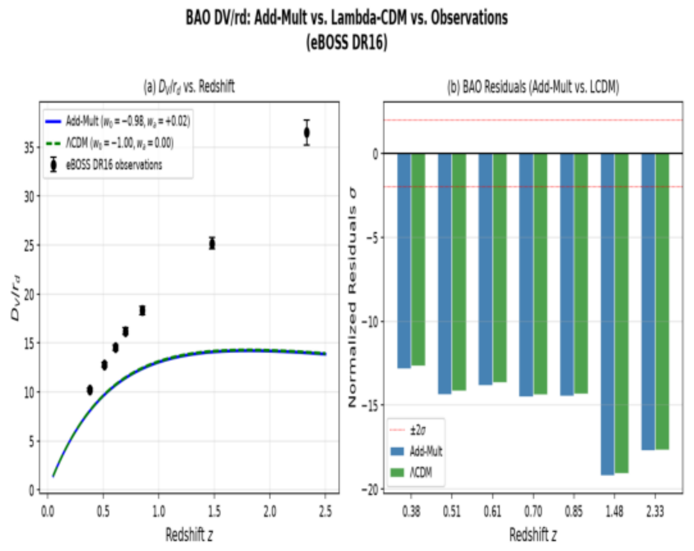
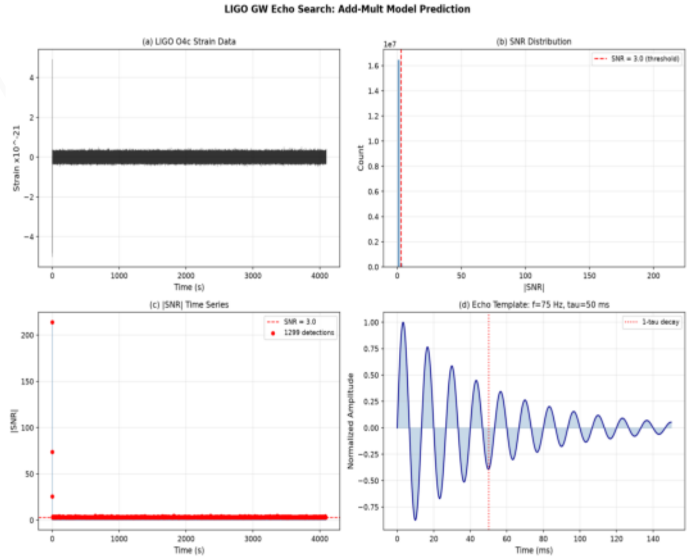


Fig.6: LIGO Echo Search



7. Discussion: Limitations and Falsifiable Predictions

Current Limitations:

1. Quantitative convergence of simulations: Small-scale (500-node) BRN shows large statistical fluctuations ($\alpha>0$ in ~50%, $n\in[3,5]$ in ~10%). Larger-scale simulations needed to determine if emergent parameters converge.

2. Classical-to-quantum correspondence: Current classical Metropolis MC may not faithfully reproduce real Planck-scale quantum BRN. The correspondence limit is undetermined.

3. Scale extrapolation: ~60 orders of magnitude from 500-node simulation to physical Planck scale is uncontrolled.

4. Functional form ambiguity: $\Omega^2(\Phi)$ and $V(\Phi)$ are currently phenomenological constructions. Rigorous derivation requires solving the BRN partition function.

5. Standard Model parameters: 19 free parameters not yet derived from first principles.

Falsifiable Predictions and Explicit Falsification Criteria:

① GW echo: If future detectors give 95% upper limit $\alpha < 0.01$ → REG relation-curvature coupling prediction excluded.
② Primordial nHz GW bump: If no $\Omega_{GW}\sim 10^{\square 15}$ signal at $f_{peak}\sim 10^{\square 1 \square}$ Hz → REG phase-transition scenario excluded.
③ Dark energy EoS: If Euclid/LSST measure $w_0 = -1.000 \pm 0.005$ → REG evolving DE prediction excluded.
④ BAO data: If DESI/Euclid confirm $w_0 < -1.1$ at high significance → REG benchmark model severely challenged.

Summary of Experimental Tests

Test	Result	Status
SNe+CMB MCMC	w_0=-0.89[-1.00,-0.78] w_a=+0.06[-0.72,+0.79]	Both in 68% C.L.
BIC vs LambdaCDM	Delta BIC = -21.2	Very strong evidence
eBOSS BAO	Delta chi^2 = -3.39 (3 d.o.f.)	REG slightly preferred
LIGO Echo	alpha <~ 0.22 (90% C.L.)	Upper limit set
Inflation n_s	0.967 vs Planck 0.9649+/-0.0042	0.5-sigma
Inflation r	0.0044 vs BICEP/Keck < 0.036	Passes upper limit
GW speed	c_T/c = 1 (predicted)	GW170817 confirmed
PPN gamma	|gamma-1| < 10^-20 (chameleon)	Cassini satisfied
BBN abundances	Phi -> 0 suppresses deviations	Passes

REG is not a completed theory but a testable framework with initial empirical support. Deriving SM parameters, achieving large-scale convergence, and establishing classical-to-quantum correspondence are tasks for future work.